

JOHN O'SULLIVAN

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EDUCATION

University of Oxford

October 2020 - March 2025

DPhil in Materials. Funded by an EPSRC Doctoral Training Partnership.

Thesis title: *Graphene transparent conductors for tandem photovoltaic cells*. Supervisor: Prof Ruy Sebastian Bonilla.

Developed novel graphene doping techniques using electrostatically charged dielectric nanolayers. The doped layers were evaluated as a transparent conducting electrode for high-efficiency tandem photovoltaic devices.

University College London

September 2016 - June 2020

MSci Physics and Molecular and Cell Biology. 1st Class Honours Thesis title: *Hybrid 2D materials for thermoelectric applications*. Supervisor: Prof Christopher A. Howard.

RESEARCH EXPERIENCE

University of Oxford, Department of Materials

Postdoctoral Researcher

April 2025 - Present

Developing Materials for Tandem Solar Cells:

In my research, I develop new techniques to fabricate sustainable materials for solar cells. In particular, I am developing a variation on a deposition technique known as atomic layer deposition (ALD), which is used to fabricate high-quality thin nanolayer films on the surfaces of materials for solar cells, and in the integrated circuits industry for computing. Through examining the chemical reactions occurring as the nanolayer forms, I develop films with properties optimised for tandem PV.

Expertise in Semiconductor and PV The fabrication of the films in particular conditions using ALD requires deep understanding of the chemistry of the materials involved, while studying the fabricated films using advanced characterisation requires an understanding of the underlying physics that govern materials properties. Finally, I exploit mathematical modelling using high-level coding languages to understand the mechanisms at play. My work in this area has been published in peer reviewed journals, and I have presented my work in international conferences.

Specialist Transparent Conductor Knowledge: Presented work on transparent conductors at several international conferences, including three oral presentations. Work selected as the **top 5%** of all contributions to the 8th World Conference on Photovoltaic Energy Conversion, where I was invited to publish in a special issue article for the high-impact *Progress in Photovoltaics: Research and Applications* (among just three other PhD students).

University of Oxford, Department of Materials

DPhil Student

October 2020 - Present

Impact and Novelty: Developed entirely novel techniques to manipulate the conductivity of graphene using charged dielectric nanolayers. Graphene doping techniques developed include depositing corona charge on dielectric membranes and polymer layers, embedding alkali ions in thin films and glass substrates, and the encapsulation of graphene with charged dielectric layers deposited by physical vapour deposition and atomic layer deposition. Fabricated films were characterised using

techniques including the Van der Pauw method, field-effect transistor measurement, kelvin probe spectroscopy, and Raman spectroscopy.

Knowledge Transfer and Publications: Presented at several international conferences, including three oral presentations. Work selected as the **top 5%** of all contributions to the 8th World Conference on Photovoltaic Energy Conversion, where I was invited to publish in a special issue article for the high-impact *Prog. Photovolt.: Res. Appl.* (among just three other PhD students).

UCL Department of Physics & Astronomy
September 2019 - June 2020

MSci Student

Investigated carbon-nanotube-2D Bi₂Te₃ hybrid material for flexible thermoelectric applications. Conducted glovebox synthesis of 2D materials dispersions, and characterised via Raman spectroscopy and X-ray diffraction.

Deutsches Elektronen-Synchrotron (DESY), Hamburg
July 2019 - September 2019

DESY Summer Student

Used Focused Ion Beam/Scanning Electron Microscopy to mill microchip samples, and reconstructed in 3D to provide elemental information.

Center for Hybrid Nanostructures (CHyN), Universität Hamburg DAAD Research Internship
June 2018 - September 2018

Analysed neural electrophysiological activity on printed biopolymer-coated nanowire substrates using the patch-clamp method.

PUBLICATIONS

O'Sullivan, J. et al. "Transparent conducting electrodes for perovskite-silicon tandem solar cells." *Nat. Rev. Phys.* 1–21 (2026). **5-year impact factor: 53.2.**

O'Sullivan, J. et al. "Towards a graphene transparent conducting electrode for perovskite/silicon tandem solar cells." *Prog. Photovolt. Res. Appl.* 31, 1478–1492 (2023).

Guo, X., Seok, H.-J., Quinn, E. L., Sharpe, M. K., McAleese, C. D., Huang, Y.-T., Li, X., Li, K., Chang, C.-Y., Wang, Y., **O'Sullivan, J.**, Moore, K. L., Ducati, C., Bonilla, R. S., Kim, H.-K., Sekkat, A., and Hoyer, R. L. Z. "Rapid Atmospheric Vapor Deposition of H:In₂O₃ Transparent Conducting Oxide Thin Films." *arXiv* 2605.16166 (2026).

Tao, X., **O'Sullivan, J.**, Wan, K., Bilotti, E., Bonilla, R. S., et al. "Roll-to-Roll Manufacturing of Flexible Thermoelectric Semiconductors Using Low-Level Direct-Current Sputtering." *Adv. Mater. Technol.* 11, 8 (2026).

Wang, Y. et al. "Impact of Precursor Dosing on the Surface Passivation of AZO/AlO_x Stacks Formed Using Atomic Layer Deposition." *Energy Adv.* 4, 553–564 (2025).

Xiong, Y. et al. "Self-aligned Laser Opening and Stencil Metallization for Silver-Free Contacts in Silicon Solar Cells." *APL Electron. Devices* 1, 2 (2025).

Sheath, B. C., Xu, X., Manuel, P., Hadermann, J., **O'Sullivan, J.**, Bonilla, R. S., Batuk, M., and Clarke, S. J. "Structures and Magnetic Ordering in Layered Cr Oxide Arsenides Sr₂CrO₂Cr₂OAs₂ and Sr₂CrO₃CrAs." *Inorg. Chem.* 61, 12373–12385 (2022).

Harberts, J., Kusch, M., **O'Sullivan, J.**, Zierold, R., and Blick, R. H. "A Temperature-Controlled Patch Clamp Platform Demonstrated on Jurkat T Lymphocytes and Human Induced Pluripotent Stem Cell-Derived Neurons." *Bioengineering* 7, 46 (2020).

SELECTED RESEARCH SKILLS

Electrical Characterisation: High temperature I-V characterisation; Van der Pauw, Hall effect and Field Effect Transistor measurements; Four-Point Probe, Capacitance-Voltage.

Materials Fabrication and Patterning: Plasma-enhanced chemical vapour deposition; atomic layer deposition; Graphene wet transfer; reactive-ion etching; thermal evaporation; spin-coating; photolithography; laser lithography of 2D materials; laser-design of metal masks; argon glovebox synthesis of 2D materials. Expertise in maintenance and repair of high vacuum systems, including turbomolecular, diffusion and rotary vane pumps.

Microscopy: Scanning Electron Microscopy; Transmission Electron Microscopy.

Spectroscopy: Kelvin Probe; Spectrophotometry; X-ray absorption spectroscopy (Diamond Light Source beamline), Ellipsometry, Raman spectroscopy; X-ray Diffraction; Energy Dispersive X-Ray Spectroscopy.

Computation: ATHENA, ARTEMIS, MATLAB; Python; Origin; Blender; Avizo, SCAPS.

OUTREACH AND TEACHING EXPERIENCE

Department of Physics, University of Oxford
September 2024 - Present

Laboratory Demonstrator

Department of Materials, University of Oxford
October 2021 - Present

Outreach Ambassador

Outreach: Delivered demonstrations and basic laboratory practicals to prospective students from disadvantaged backgrounds. Met with prospective applicants to offer a student's point of view and share my opinions and experience of studying Materials Science.

Junior Demonstrator

October 2021 - October 2022

Facilitated the smooth running of laboratory practicals for groups of 20-30 undergraduate students.

Centre for Talented Youth, Ireland (CTYI)
June 2020 - July 2020

Teaching Instructor

Designed the syllabus, conducted review quizzes, provided academic feedback and wrote a 70-page textbook to support an intensive course on the physics of aviation.

UCL Department of Physics & Astronomy
December 2018 - June 2020

Student Ambassador

AWARDS AND ACHIEVEMENTS

2026: ESRF (European Synchrotron Radiation Facility), XMaS Beamline, Awarded 4 days of Operando X-ray absorption spectroscopy of ZnO structural changes under damp-heat conditions with applied bias and controlled gas atmospheres.

2024: EPSRC IAA Doctoral Prize (indicative of being in the top 5-10% of my PhD cohort).

2022, 2023: Mansfield College Conference Travel Fund (£800).

2022: SuperSolar International Conference Travel Award (£500).

2022: Graphene 2022 Student Travel Grant (€150).
 2021: Roberts' Skills Training Fund (£500).
 2020: Winner of the Society of Chemical Industry's *Bright SCIdea* competition (£5090).
 2020: EPSRC Doctoral Training Partnership PhD Scholarship.
 2019: Deutsches Elektronen-Synchrotron (DESY) Science Research Internship Scholarship.
 2018: DAAD Research Internship in Science and Engineering Scholarship.
 2016: Roibeárd Ó'Buachalla Medal. For achieving among the highest exam results nationally.

CONFERENCE PRESENTATIONS

- Tandem PV Workshop (Berlin, Germany) 2026: *Design Considerations for In-free Transparent Electrodes for Perovskite-Silicon Tandem Solar Cells*, Poster.
- Photovoltaics Science, Applications and Technology Conference (Newcastle, UK) 2026: *Transparent Electrodes for Perovskite-Silicon Tandem Solar Cells*, Oral.
- Tandem PV Workshop (Hasselt, Belgium) 2025: *Ion-Charged Dielectric Doped Graphene as a Transparent Conductor for Tandem Cells*, Oral.
- 2024 Oxford Solar Research Symposium (Oxford, UK) 2024: *Electrostatically doped graphene as a novel transparent conducting electrode for tandem PV cells*, Oral.
- Photovoltaic Science and Engineering Conference (Shenzhen, China) 2023: *PMMA-Doped Graphene Transparent Conducting Electrodes for Tandem Solar Cells*, Poster.
- Photovoltaics Science, Applications and Technology Conference (London, UK) 2023: *Ion-Charged Dielectric Doping of Graphene for Indium-Free Tandem Photovoltaics* Oral.
- Transition Metal Dichalcogenides Conference 2023 (Cambridge, UK) 2023: *Ion-Charged Dielectric Polymers for Electrostatic Doping of 2D Materials*, Poster.
- Institute of Physics Advances in Photovoltaics: Tandem and Bifacial Photovoltaics 2022 (London, UK): *Towards a Graphene Transparent Conducting Electrode for Perovskite/Silicon Tandem Solar Cells*, Oral.
- 8th World Conference on Photovoltaic Energy Conversion 2022 (Milan, Italy): *Towards a Graphene Transparent Conducting Electrode for Perovskite/Silicon Tandem Solar Cells*, Oral.
- Graphene 2022 (Aachen Germany): *Electrostatic Field-Effect Doping of Graphene Using Ion-Charged Dielectrics*, Poster.